

Demographics and Housing Booms

Yun Gu Cho, Jinho Kim, Hyunseo Park, Jaey Yoo

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Abstract

This paper studies whether demographic age structure is associated with housing-boom vulnerability. We define large housing booms as one-standard-deviation positive deviations of real house prices from location-specific trend. In an OECD panel of 29 countries over 1970–2021 and a U.S. county panel over 1975–2024, locations with higher young-to-old population ratios are more likely to experience large housing booms. In the country and time fixed-effects specification, a one-unit increase in the young-to-old ratio is associated with an 18.6 percentage-point higher boom probability; in the corresponding U.S. county specification, the estimate is 4.3 percentage points. HMDA mortgage data further show that young applicants increase mortgage-market participation disproportionately in boom county-years, with similar results for loan originations. To rationalize these facts, we study a six-generation housing OLG model with tenure choice, gradual price adjustment, adaptive house-price-growth beliefs, and age-specific income profiles. The model shows that a young economy can generate a larger house-price response to the same housing-preference shock when more households are located near the life-cycle tenure-adjustment margin.

Keywords: housing booms; demographics; OLG model; life-cycle housing demand

1 Introduction

The relationship between demographics and housing markets has long been an important topic in macroeconomics and housing economics. A substantial body of work documents that demographic composition is a key determinant of the level and trend of housing demand (Mankiw and Weil, 1989; Bakshi and Chen, 1994; Takáts, 2012; Nishimura and Takáts, 2012). However, much less is known about whether demographic structure also shapes the likelihood and size of housing booms. While aging is an ongoing issue in most developed economies, research on housing booms has focused primarily on credit conditions, monetary policy, collateral constraints, and speculative demand. Standard macroeconomic models of housing markets generate house-price fluctuations through financial frictions and aggregate shocks, but typically abstract from age heterogeneity among households.

This paper provides evidence that demographic age structure is systematically associated with housing-boom incidence and that this pattern is consistent with a life-cycle housing-demand mechanism. Using data at both the OECD country level and the U.S. regional level, we show that locations with relatively younger populations are more likely to experience large upward deviations of house prices from trend. We then ask whether this empirical pattern can be reproduced inside a stylized life-cycle housing model. The model exercise is not a quantitative estimation of any particular country or county; it is a disciplined mechanism exercise. Holding the housing-preference shock fixed, can a younger economy generate a larger house-price boom response than an older economy?

Our empirical analysis begins by documenting the aggregate relationship between age structure and boom incidence. We define a large housing boom as an episode in which the cyclical component of real house prices exceeds one standard deviation above its location-specific trend. Using an OECD country panel covering 29 countries from 1970 to 2021 and a U.S. county-year panel over 1975–2024, we show that locations with a higher ratio of young adults to older adults are more likely to experience large upward deviations of house prices from trend.

The magnitudes are economically meaningful. In the country panel, the most demanding linear

probability model specification implies that a one-unit increase in the young-to-old population ratio is associated with an 18.6 percentage-point increase in the probability of a large housing boom. In the U.S. county panel, the county and time fixed-effects specification implies a 4.3 percentage-point increase. Because the ratio is not standardized, these one-unit effects should be read as slope estimates rather than as typical demographic shifts. The results establish a robust aggregate demographic gradient: higher young-to-old ratios are associated with higher boom incidence.

We next examine the household-level demand margin underlying this aggregate pattern. Although the young-to-old ratio evidence shows that younger locations have higher boom incidence, it could in principle reflect other location-level forces correlated with age structure. A natural implication of the life-cycle housing-demand view is that, conditional on a local boom, younger households should enter the mortgage market more aggressively than older households. We therefore use Home Mortgage Disclosure Act (HMDA) mortgage application data to test whether young applicants increase mortgage-market activity disproportionately when local house prices are booming. The evidence supports this implication: in boom county-years, young applicants exhibit application intensity 4.3–6.6 percent higher than older comparison groups when scaled by the older groups' boom-period mean, and 5.3–8.0 percent higher when scaled by their full-sample mean. We interpret this as mechanism-consistent evidence on age-specific participation during boom periods rather than as a fully causal estimate of boom onset on young households. These results are robust to alternative definitions of age groups, application outcomes, stricter boom thresholds, and basic sample restrictions.

The second part of the paper develops this model-based mechanism exercise. The model contains six overlapping generations that differ in age, income, asset holdings, tenure status, and survival probabilities. Households choose consumption, savings, and whether to own or rent. A housing-preference shock raises the utility value of ownership, and house prices adjust gradually to excess demand. This environment allows us to examine whether the same shock can produce different peak house-price responses in young and old demographic structures.

The key ingredient is intergenerational income heterogeneity. In the model, deterministic income follows a hump-shaped life-cycle profile: younger households have relatively low income, income peaks in middle age, and then declines among older households. This profile creates a tenure-adjustment margin that is absent in a purely mechanical demographic reweighting exercise. When a preference shock raises the value of homeownership, the response is strongest among cohorts with enough income and assets to move into ownership but not yet fully saturated in housing demand. A younger economy places more mass near this margin. As a result, the young economy generates a higher peak house-price response than the old economy.

In the baseline simulation, the same housing-preference shock raises house prices by 15.6 percent at the peak in the young economy and by 13.7 percent in the old economy. Thus, a stylized housing OLG model can qualitatively reproduce the central empirical pattern once it includes both demographic structure and age-specific income differences. The result does not require assuming that young households are intrinsically more speculative; instead, demographics matter because they determine how many households sit near the life-cycle housing-demand margin when the shock arrives.

This paper is related to three strands of literature. First, the paper contributes to the literature on demographics and housing markets. From a life-cycle perspective, age structure shapes household formation, saving behavior, portfolio allocation, and the demand for owner-occupied housing (Bakshi and Chen, 1994). The classic “baby boom” hypothesis of Mankiw and Weil (1989) argues that changes in cohort size generate large shifts in housing demand and house prices. Subsequent work shows that population aging can affect house prices and asset valuations over long horizons (Takáts, 2010, 2012; Nishimura and Takáts, 2012; Gevorgyan, 2019). This literature establishes that demographics are a powerful determinant of long-run housing demand and valuation. Our paper shifts the focus from long-run price levels to boom vulnerability. We show that demographic composition also predicts the likelihood of large upward deviations of house prices from trend.

Second, the paper contributes to the literature on housing booms, credit constraints, and macro-financial amplification. A large literature studies housing booms as unusually large deviations of

real house prices from trend and emphasizes the macro-financial risks associated with asset-price booms (Borio and Lowe, 2002; Detken and Smets, 2004; Goodhart and Hofmann, 2008; Jordà et al., 2015). Our analysis builds on this approach by asking why some locations are more prone to large housing booms than others. We show that demographic composition is a robust predictor of boom incidence across both national and local housing markets.

Third, the paper contributes to quantitative life-cycle and OLG approaches to housing demand. By embedding tenure choice, demographic weights, gradual price adjustment, and age-specific income profiles in a six-generation housing model, we show that the effect of demographics on boom amplification depends on more than the mechanical share of young households. The model clarifies why intergenerational income differences are important for reproducing the empirical pattern: they determine which cohorts are close to the owner-renter margin and therefore how strongly aggregate demand responds to a common preference shock.

The rest of the paper proceeds as follows. Section 2 presents the empirical evidence on demographics and large housing booms. Section 3 develops the housing OLG model and uses it to assess whether the empirical pattern can be rationalized by a life-cycle housing-demand mechanism. Section 4 concludes.

2 Effects of Demographics on Housing Booms

2.1 Data

Our empirical analysis combines several datasets at both the country and regional levels. For the cross-country analysis, we use OECD house price indicators, OECD population data by age group, and real GDP per capita for 29 countries over the period 1970–2021. For the U.S. county analysis, we use annual FHFA county house price indexes over 1975–2024, together with county population data and housing supply elasticity estimates from Saiz (2010). The number of county-year observations differs across specifications because housing-price availability and the availability of the supply-elasticity control are not identical across counties.

Table 1: Summary of Datasets

Analysis	Source	Unit	Sample	Variables
Cross-country	OECD	Country $\times$ year	1970–2021	Real house prices, real GDP per capita, population by age group
U.S. county	FHFA, U.S. Census	County $\times$ year	1975–2024	Real house prices, population by age group, supply elasticity

Notes: For the U.S. regional analysis, housing supply elasticity is from Saiz (2010).

2.2 Large Housing Booms

We define a large housing boom as an episode in which real house prices rise substantially above their long-run trend. This definition builds on the literature on housing booms and bubbles. For example, Borio and Lowe (2002), Detken and Smets (2004), and Goodhart and Hofmann (2008) define housing booms as periods in which real house prices rise sufficiently above their HP-filtered trend under relatively high smoothing parameters. Jordà et al. (2015) identify housing bubble episodes using two conditions: the cyclical component of real house prices must exceed one standard deviation, and house prices must subsequently decline by at least 15 percent within three years.

In this study, we follow Jordà et al. (2015) in using the HP filter (Hodrick and Prescott, 1997) to extract the cyclical component of real house prices, but we omit the subsequent price-decline condition. By dropping this ex-post crash requirement, we retain episodes in which house prices rise sharply above trend, regardless of whether a large decline materializes within a specific window.

This modification is motivated by our different research objective. Whereas the definition in Jordà et al. (2015) is designed to identify boom-bust episodes and study the macro-financial consequences of bursting housing bubbles, our focus is on the incidence of large upward house-price deviations associated with demographic structure. Requiring an ex-post price collapse would condition the boom definition on a subsequent outcome that may depend on factors outside the demographic mechanism, such as credit conditions, policy responses, housing supply, or monetary tightening. We therefore retain the price-elevation criterion but omit the subsequent crash requirement, so that our boom indicator captures large upward deviations in real house prices rather than realized boom-bust episodes.

We decompose real house prices into trend and cyclical components using the HP filter:

$$p_{i,t} = \tau_{i,t} + c_{i,t}, \quad (1)$$

where $p_{i,t}$ denotes the log real house price in location i at time t , $\tau_{i,t}$ is the trend component, and $c_{i,t}$ is the cyclical component. Because both the country and county panels are annual, we set the HP smoothing parameter to $\lambda = 100$.

We then define the large housing boom indicator as

$$\text{LHB}_{i,t} = \begin{cases} 1, & \text{if } c_{i,t} > \sigma_{c_i}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where σ_{c_i} is the standard deviation of $c_{i,t}$ for location i over the sample period. Thus, a large housing boom is identified whenever the cyclical component of real house prices exceeds one standard deviation of the location-specific cyclical component. Because the HP trend and σ_{c_i} are constructed from the available sample, the indicator should be interpreted as an ex-post classification of large upward deviations rather than as a real-time forecast signal.

2.3 Young-to-Old Population Ratio

The literature does not adopt a single age cutoff for distinguishing young, middle-aged, and old cohorts. The relevant boundaries vary across studies depending on the question of interest. In the housing literature, age profiles are typically motivated by life-cycle patterns in household formation, housing demand, and asset accumulation. The demographic literature similarly uses alternative cohort definitions depending on the mechanism under consideration.

We define the young cohort as individuals aged 20–39 and the older cohort as individuals aged 50–69. The numerator is intended to capture younger adults who are entering the housing market and forming housing demand, whereas the denominator captures older households who are, on average, more likely to have accumulated housing wealth and to adjust housing consumption or

rebalance their portfolios. Accordingly, our measure should be interpreted not as a conventional old-age dependency ratio, but as a cohort-balance measure designed to capture life-cycle patterns in housing demand and asset holdings.

We define the young-to-old population ratio as

$$YO_{i,t} \equiv \frac{\text{Population}_{i,t}^{(20-39)}}{\text{Population}_{i,t}^{(50-69)}}, \quad (3)$$

where a higher value of $YO_{i,t}$ indicates a relatively larger young cohort compared with the older cohort. In the Appendix, we examine the robustness of the results to alternative age-group definitions.

2.4 Descriptive Evidence

Before turning to regression analysis, we present descriptive evidence on the relationship between demographic structure and large housing booms. Figure 1 plots the probability of a large housing boom against the young-to-old population ratio in the country panel and the U.S. county panel. Panel (a) displays the country-level relationship, while Panel (b) shows the analogous relationship at the U.S. county level.¹

The country-level plot reveals a clear positive association: countries with a higher young-to-old ratio tend to experience large housing booms more frequently. The U.S. county-level plot shows a similar positive relationship, although the association is weaker than in the country panel. This difference is not surprising given the finer geographic resolution and greater heterogeneity of local housing markets. These patterns are descriptive, but they are consistent with the hypothesis that younger demographic structures are associated with more frequent large housing booms and motivate the regression analysis that follows.

¹The country-level plot is reported without population weights because each country is treated as a national housing-market regime, with its own macroeconomic, institutional, and mortgage-market environment. The U.S. county-level plot, by contrast, uses county population weights. Counties differ substantially in market size, and treating very small and very large counties symmetrically can give disproportionate influence to idiosyncratic demographic variation in small local markets. Because the proposed mechanism operates through household demand and local housing-market participation, population provides a natural proxy for the size of the relevant housing market. Appendix Figure 3 reports the corresponding unweighted county-level bin-scatter plot; the relationship is weaker but remains positive.

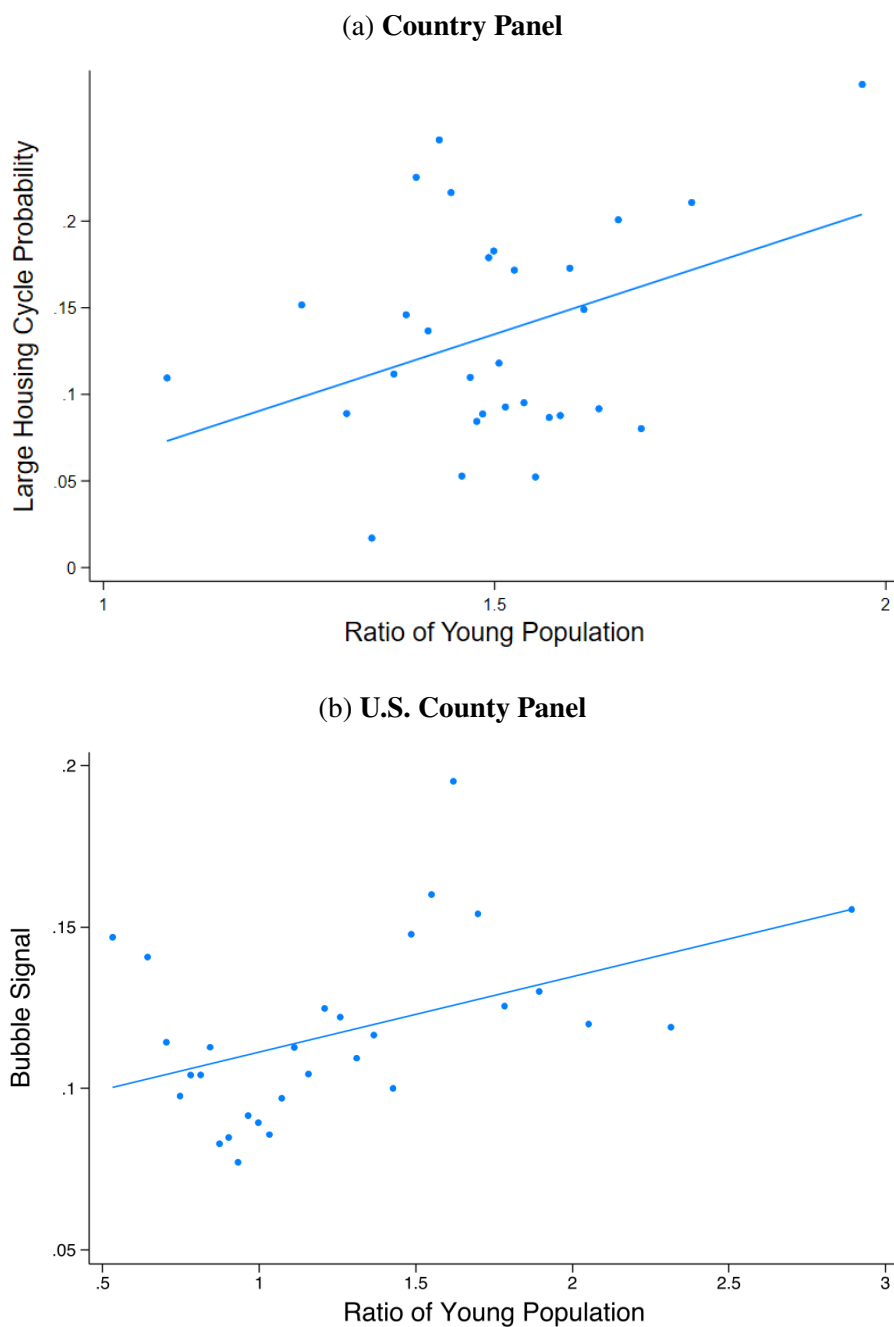


Figure 1: Bin Scatter Plot of Large Housing Booms and Young-to-Old Population Ratio

Notes: This figure presents bin-scatter plots of the large housing boom probability against the young-to-old population ratio. Panel (a) shows the country-level result and Panel (b) shows the U.S. county-level result. Observations are grouped into 30 equal-sized bins based on the young-to-old ratio, and each plot displays the mean large housing boom probability within each bin along with a linear fit.

2.5 Demographics and Large Housing Booms

We estimate a sequence of linear probability model (LPM) specifications to examine how the young-to-old population ratio is associated with the probability of a large housing boom. In both the cross-country and U.S. county analyses, we begin with parsimonious specifications and then progressively introduce additional controls and fixed effects in order to assess the robustness of the estimated relationship.

Our baseline specification is given by

$$\text{LHB}_{i,t} = \beta_0 + \beta_1 \text{YO}_{i,t} + Z'_{i,t} \phi + \alpha_i + \gamma_t + \varepsilon_{i,t}, \quad (4)$$

where $\text{LHB}_{i,t}$ is an indicator equal to one if location i experiences a large housing boom at time t , $\text{YO}_{i,t}$ is the young-to-old population ratio, $Z_{i,t}$ is a vector of control variables, α_i denotes location-specific fixed effects when included, and γ_t denotes time fixed effects when included. Standard errors are clustered at the location level throughout.

In the cross-country analysis, the control vector includes log real GDP per capita, while in the U.S. county analysis it includes housing supply elasticity. Across specifications, we progressively add controls, location fixed effects, and time fixed effects. Because the LPM is linear, the coefficient β_1 is directly interpretable as the change in the probability of a large housing boom associated with a one-unit increase in the young-to-old ratio, holding other covariates constant. This approach allows us to assess whether the relationship between demographic structure and large housing booms is robust to various specifications without the complications arising from nonlinear models.

Tables 2 and 3 report the results. Across both the cross-country and U.S. county samples, the coefficient on the young-to-old ratio is positive and statistically significant in all specifications, indicating that locations with younger demographic structures have higher incidence of large upward deviations of house prices from trend.

Table 2 shows a strong and robust relationship in the country panel. The coefficient on the young-to-old ratio is positive in all five specifications, ranging from 0.045 in the most parsimonious

specification to 0.231 in the specification with country fixed effects and GDP per capita. In the most demanding specification with both country and time fixed effects (column 5), a one-unit increase in the young-to-old ratio is associated with an 18.6 percentage-point increase in the probability of a large housing boom. Because this variable is a ratio rather than a standardized regressor, the one-unit coefficient is best interpreted as the slope of the boom-probability relationship. The estimated relationship remains statistically significant after introducing country and time fixed effects, suggesting that the result is not driven solely by time-invariant cross-country heterogeneity or common global shocks. GDP per capita enters positively and is statistically significant at the 5% level in both specifications in which it is included, indicating that higher income levels are also associated with a greater likelihood of large housing booms, though the demographic composition variable remains the more robust predictor.

Table 2: Linear Probability Model: Country Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.045*	0.160***	0.231***	0.147**	0.186**
	(0.023)	(0.030)	(0.052)	(0.064)	(0.077)
GDP per Cap.			0.083**		0.323**
			(0.039)		(0.118)
Observations	1,262	1,262	1,225	1,262	1,225
Country FE	No	Yes	Yes	Yes	Yes
Time FE	No	No	No	Yes	Yes
R^2	0.0045	0.0252	0.0298	0.1895	0.2058

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the country level.

Table 3 shows that the same qualitative relationship holds in the U.S. county panel. The coefficient on the young-to-old ratio is positive and highly significant across all specifications, ranging from 0.013 in the time-FE-only specification (column 3) to 0.075 in the pooled specification (column 1). In the specification with both county and time fixed effects (column 5), a one-unit increase in the young-to-old ratio is associated with a 4.3 percentage-point increase in the probability of a large housing boom. Supply elasticity enters with a negative coefficient in column (4), consistent with the view that more elastic housing supply dampens the translation of demand shifts into house

price fluctuations, though the estimate is not statistically significant. The substantial rise in R^2 in specifications with time fixed effects suggests that common temporal shocks explain an important share of boom incidence in U.S. counties, although the demographic effect remains statistically meaningful.

The difference in the estimated magnitude of the young-to-old population ratio effect across the country and U.S. county panels may reflect several factors. One possible explanation is that the country panel reflects slow-moving differences in national demographic regimes, housing-finance institutions, and macroeconomic conditions. In contrast, the county panel exploits within-country variation across local housing markets, where population mobility, commuting patterns, and idiosyncratic shocks can weaken the mapping from age structure to local housing booms. More broadly, the comparison suggests that while the magnitude of the coefficients differs across samples, the positive relationship between the young-to-old ratio and the likelihood of a large housing boom remains robust across very different empirical settings.

Table 3: Linear Probability Model: U.S. County Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.075*** (0.001)	0.044*** (0.006)	0.013*** (0.004)	0.022*** (0.006)	0.043** (0.019)
Supply Elasticity				-0.002 (0.002)	
Observations	101,567	101,567	101,567	37,205	101,567
County FE	No	Yes	No	No	Yes
Time FE	No	No	Yes	Yes	Yes
R^2	0.1178	0.0159	0.2887	0.3198	0.3046

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by county population. Standard errors are clustered at the county level. Column (4) includes housing supply elasticity; column (5) reports the county and year fixed-effects specification without the elasticity control.

2.6 Young Applicants and Mortgage-Market Participation During Booms

The previous regression results connect the two ends of the proposed mechanism: locations with younger demographic structures have higher boom incidence. They do not, however, identify the

intermediate demand margin through which demographics may translate into larger housing booms.

To examine this margin directly, we turn to mortgage application data from the Home Mortgage Disclosure Act (HMDA). HMDA provides loan-level data on mortgage applications and their outcomes for covered financial institutions. The data span a broad segment of the U.S. mortgage market and include information on loan purpose, property location, applicant characteristics, and applicant age in decade-based bins.² This makes HMDA a natural source for measuring the extensive-margin demand for housing finance: an application reflects a household’s attempt to obtain mortgage credit, while originated loans provide a narrower measure of completed housing-finance transactions.

We construct a county–age-group–year panel over the 2018–2024 HMDA sample period, which is the period over which applicant age is reported in the public HMDA files. The outcome is mortgage applications per 1,000 age-specific population in county c , age group a , and year t . We identify housing boom county-years using the same large-boom definition as in the country and county analyses above: the cyclical component of real house prices must exceed one standard deviation above its location-specific trend. We then estimate the following triple-difference specification:

$$Y_{c,a,t} = \beta (\text{Boom}_{c,t} \times \text{Young}_a) + \alpha_{c,a} + \gamma_{t,a} + \delta_{c,t} + \varepsilon_{c,a,t}, \quad (5)$$

where $Y_{c,a,t}$ denotes applications per 1,000 age-specific population. The fixed effects are county-by-age effects $\alpha_{c,a}$, year-by-age effects $\gamma_{t,a}$, and county-by-year effects $\delta_{c,t}$. The specification is highly saturated: the county-by-year fixed effects absorb all county-year shocks and any level effect of the boom indicator itself, including local credit conditions, local house-price growth, and other time-varying county-level factors common across age groups. Identification therefore comes from whether, within the same county-year, young applicants increase mortgage-market activity more than older applicants when that county is in a boom. The coefficient β measures the differential increase in young applicants’ application intensity in boom county-years relative to the older comparison

²Estimates from the Consumer Financial Protection Bureau (CFPB) indicate that HMDA captures about 92 percent of mortgage originations nationally (2017). Estimates from the Federal Reserve Bank of Minneapolis suggest coverage above 90 percent in metropolitan counties and about 80 percent in remote nonmetropolitan counties (2015).

group. We interpret this specification as evidence on within-county-year age-group differences in mortgage-market participation, rather than as a fully causal estimate of the effect of boom onset on young households.

Table 4 reports the results. Across columns (1)–(4), the estimated boom-period young-minus-old differential ranges from 1.00 to 1.69 additional mortgage applications per 1,000 age-specific population. The natural denominator for interpreting these coefficients is the comparison group’s application intensity in boom county-years, because the regression coefficient is the incremental young-versus-old gap during boom episodes. Relative to the older comparison groups’ mean boom-period application intensity, these estimates correspond to roughly 4.3% to 6.6% higher application intensity for young groups. Using the older groups’ full-sample mean application intensity instead yields a somewhat larger range of about 5.3% to 8.0%. These results indicate that young applicants disproportionately increase their mortgage-market activity during housing booms relative to older applicants. The finding complements the LPM evidence by showing not only that younger areas have higher boom incidence, but also that young cohorts are themselves more active participants in the mortgage market precisely when booms occur. This pattern is consistent with the life-cycle interpretation that younger households, who are at the stage of household formation and initial home purchase, respond more strongly to boom-period conditions.

One potential concern is that the increase in mortgage applications by younger applicants does not necessarily reflect actual housing demand. For example, younger households may submit more applications per property by shopping across multiple lenders, or their applications may be less likely to translate into completed home purchases. While HMDA records only applications that enter the lender’s formal underwriting process—that is, oral or written requests submitted according to the institution’s application procedures, rather than informal information inquiries—it is nonetheless useful to examine whether our results are robust to alternative outcomes and comparison groups.

To address this concern, we re-estimate our baseline specification using alternative outcome measures: originated loans and denied applications. Table 5 reports the results. When restricting the outcome to originated loans, the estimated coefficients decline in magnitude but remain large and

Table 4: HMDA Baseline: Application Intensity in Boom County-Years

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Comparison group	45–74	45–74	55–74	55–74
$\text{Boom}_{c,t} \times \text{Young}_a$	1.002*** (0.179)	1.346*** (0.227)	1.335*** (0.235)	1.686*** (0.264)
Observations	76,328	95,587	57,161	76,420
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	8,349	11,113
R^2	0.8896	0.8755	0.8991	0.8798

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (5) using the indicated young and comparison groups. Columns (3)–(4) use an older comparison group as a robustness check. Regressions are weighted by age-specific population. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

statistically significant. This stricter outcome indicates that the increased application activity among young applicants during booms translates into higher realized borrowing, rather than reflecting only multiple applications or unsuccessful credit searches. Denied applications also rise significantly for young applicants during boom periods, suggesting that part of the increased activity reflects a broader expansion in credit-seeking behavior. Quantitatively, the origination margin accounts for the bulk of the baseline application response, suggesting that the result largely reflects completed mortgage transactions rather than unsuccessful credit searches. This pattern supports the interpretation that effective housing demand by young households rises disproportionately during boom periods. Additional robustness checks are reported in Appendix Section C.2.

Beyond young households' extensive-margin response, Appendix C.3 examines whether they also amplify housing booms through an intensive-margin channel by taking on higher leverage during boom periods. The estimated effects are modest in magnitude but statistically significant: relative to older borrowers, young borrowers exhibit higher loan-to-value and debt-to-income ratios in boom county-years.

Table 5: HMDA Robustness: Mortgage Application Outcomes

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Outcome	Originations	Originations	Denials	Denials
Comparison group	45–74	45–74	45–74	45–74
Boom _{c,t} × Young _a	0.820*** (0.123)	0.968*** (0.148)	0.071*** (0.021)	0.105*** (0.024)
Observations	76,328	95,587	76,328	95,587
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	11,113	13,892

Notes: Outcomes are measured per 1,000 age-specific population. Each column estimates equation (5) using the indicated outcome and age-group definition. Regressions are weighted by age-specific population. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

3 A Housing OLG Model

We next ask whether the empirical pattern can be reproduced in a stylized life-cycle housing environment. The exercise uses a structural overlapping-generations model, but its purpose is deliberately modest. Rather than estimating the model or matching a particular country or county, we use it as a laboratory: holding fixed the housing-preference shock, can a younger economy generate a larger house-price boom response than an older economy?

The answer is yes once the model includes intergenerational income differences. Demographics alone mechanically changes population weights, but it does not by itself guarantee that a young economy has the larger price response. The important interaction is between the demographic distribution and the life-cycle income profile. When income varies across generations, cohorts differ in their ability and incentive to move into ownership after a housing-preference shock. A young economy places more mass close to this tenure-adjustment margin, so the same shock generates a larger increase in aggregate housing demand and a larger peak price response.

3.1 Model environment

The economy contains six overlapping generations, indexed by $g = 0, \dots, 5$, corresponding to the age groups 21–30, 31–40, 41–50, 51–60, 61–70, and 71–80. Demographic composition is governed by generation-specific survival probabilities. Let ρ_g denote the probability that a member of generation g survives into generation $g + 1$. Population masses are constructed as

$$N_0 = 1, \quad N_g = N_{g-1}\rho_{g-1}, \quad (6)$$

and normalized as

$$\pi_g = \frac{N_g}{\sum_{h=0}^5 N_h}. \quad (7)$$

These normalized weights determine aggregate demand. We use survival probabilities as a parsimonious device for generating two stationary demographic distributions. The young demographic scenario places more mass on younger and middle-aged cohorts, while the old scenario shifts population mass toward older cohorts.

In each period, households choose consumption, savings, and tenure status. Tenure is denoted by $s \in \{0, 1\}$, where $s = 1$ is owner-occupation and $s = 0$ is renting. Flow utility is

$$u(c_t, s_t; \Omega_t) = \frac{c_t^{1-\sigma} - 1}{1 - \sigma} + \Omega_t s_t, \quad (8)$$

where Ω_t is the housing-preference shifter. We use Ω_t for this shifter to avoid confusing it with the young-population share used elsewhere in the paper. The terminal generation consumes and leaves a warm-glow bequest, with bequest utility

$$v(B) = \phi \frac{(B + \underline{B})^{1-\sigma} - 1}{1 - \sigma}. \quad (9)$$

Households face transaction costs when buying and selling housing, pay rent when they do not own, and earn a risk-free return on financial assets. Income differs both across and within generations.

The deterministic component follows a hump-shaped life-cycle profile based on Guvenen et al. (2015), normalized across the six generations as

$$\bar{y}_g = (0.705, 1.027, 1.245, 1.240, 1.000, 0.969), \quad (10)$$

which captures lower income among young households, peak income in middle age, and lower income among older households. This intergenerational income profile is central to the model exercise because it determines which cohorts are near the owner-renter margin. Within each generation, households also draw an idiosyncratic income shock from a grid of nine equally-spaced points. The joint distribution of assets, tenure status, and income type is tracked explicitly for each cohort.

Households form expectations adaptively. Let $y_t = \log p_t - \log p_{t-1}$ denote observed house-price growth. All cohorts share a common belief $\hat{b}_t$ that follows a constant-gain Kalman updating rule,

$$\hat{b}_t = (1 - K)\hat{b}_{t-1} + Ky_t. \quad (11)$$

Each household projects future house prices according to

$$E_t p_{t+k} = p_t \exp\left(\hat{b}_t \frac{1 - \delta^k}{1 - \delta}\right), \quad \delta = 0.85. \quad (12)$$

The decay parameter δ prevents long-horizon beliefs from generating explosive price projections in the finite life-cycle problem.

For each generation, the household problem is solved by backward induction. Each household chooses tenure status $s \in \{0, 1\}$ and next-period assets by maximizing lifetime utility; tenure is selected deterministically as the argmax of the value of owning versus renting. Aggregate housing demand is the population-weighted sum of cohort-level ownership shares:

$$D_t = \sum_{g=0}^5 \pi_g \tilde{s}_{g,t}, \quad (13)$$

where $\tilde{s}_{g,t} = \int \mathbf{1}\{s^*(a, s, i; g, t) = 1\} d\mu_{g,t}(a, s, i)$ is the ownership share in cohort g , integrated over the joint distribution $\mu_{g,t}$ of assets, current tenure status, and income. House prices adjust gradually to excess demand,

$$\log p_{t+1} = \log p_t + \lambda(D_t - D^{ss}), \quad \lambda = 1.0, \quad (14)$$

where D^{ss} is the scenario-specific steady-state demand. Thus, prices are not re-cleared instantaneously each period; instead, the model generates a dynamic price-adjustment process in which demand pressure feeds into future prices and then back into beliefs.

The numerical implementation uses the parameter values summarized in Table 6. The parameters are not estimated in this exercise. They are chosen to provide a transparent model laboratory in which the same housing-preference shock can be applied to economies that differ only in demographic composition. Several elements are disciplined by existing work: the life-cycle and demographic structure follows the long tradition linking age composition to housing demand (Mankiw and Weil, 1989; Takáts, 2012); the deterministic income profile is based on Guvenen et al. (2015); the housing block follows the macro-housing tradition in which housing demand, house prices, and household balance-sheet positions interact dynamically (Iacoviello, 2005); and the belief-updating rule is a constant-gain implementation of standard linear filtering logic (Kalman, 1960; Harvey, 1989; Durbin and Koopman, 2012). The remaining numerical values, including grid sizes, the price-adjustment speed, and the shock size, should be read as transparent simulation normalizations rather than externally estimated parameters.

3.2 Simulation design

The simulation compares two demographic scenarios: a young society and an old society. The scenarios differ only in the normalized population weights generated by equation (7). For each scenario, we first compute the steady-state house price under $\Omega^{ss} = 0.10$ and zero house-price-growth beliefs. We then start each economy from its own steady state and apply the same housing-preference

Table 6: Calibration of the Housing OLG Model

Object	Value	Description
Life cycle	$G = 6$	Six age groups: 21–30, 31–40, 41–50, 51–60, 61–70, and 71–80.
Preferences	$\beta = 0.96, \sigma = 2.0$	Discount factor and coefficient of relative risk aversion.
Bequest motive	$\phi = 0.40, \underline{B} = 0.30$	Warm-glow bequest weight and bequest floor for the terminal generation.
Financial return	$r_a = 0.02$	Risk-free return on financial assets.
Rental cost	Rent = 0.06	Per-period cost paid by renter households.
Housing transaction costs	$\kappa_b = \kappa_s = 0.02$	Proportional buying and selling costs.
Housing supply	$\bar{H} = 0.55$	Fixed aggregate housing supply used to pin down the steady-state house price.
Housing preference	$\Omega^{ss} = 0.10$	Steady-state value of the owner-occupation preference shifter.
Price adjustment	$\lambda = 1.0$	Speed in the law of motion $\log p_{t+1} = \log p_t + \lambda(D_t - D^{ss})$.
Belief updating	$K = 0.618, \delta = 0.85$	Constant Kalman gain and decay parameter in projected future house prices.
Income profile	(0.705, 1.027, 1.245, 1.240, 1.000, 0.969)	Deterministic generation-specific income levels from equation (10).
Idiosyncratic income risk	$N_i = 9, \eta_i \in [-0.45, 0.45]$	Nine-point iid income shock grid with normal weights.
Asset and savings grids	$N_a = 24, a \in [0, 12]; N_f = 13, f \in [0.05, 0.95]$	Asset grid and candidate savings fractions used in the household problem.
Demographic scenarios	Young: (0.995, 0.990, 0.950, 0.880, 0.700, 0) Old: (0.998, 0.995, 0.990, 0.975, 0.940, 0)	Survival probabilities used to construct the young and old population distributions.
Preference shock	$\rho_\omega = 0.85, \varepsilon_{t^*} = 0.15, t^* = 20, T = 50$	AR(1) housing-preference shock and simulation horizon.

Notes: Values are the parameter values used in the OLG model in this paper. The Kalman gain is implied by signal and noise standard deviations equal to one. Survival probabilities are used to generate the normalized demographic weights in equation (7); all other parameters are held fixed across the young and old demographic scenarios.

shock. Because the empirical object is a large upward deviation of house prices from trend, the main model statistic is the peak log deviation of house prices from steady state.

The shock follows an AR(1) process: $\Omega_t = \Omega^{ss} + \hat{\omega}_t$ where $\hat{\omega}_t = \rho_\omega \hat{\omega}_{t-1} + \varepsilon_t$, with persistence $\rho_\omega = 0.85$ and a single innovation $\varepsilon_{t^*} = 0.15$ at period $t^* = 20$ (all other $\varepsilon_t = 0$). After the innovation, the preference shifter decays back to its steady-state value at rate ρ_ω . The simulation horizon is 50 model periods, and the reported price response is the log deviation from each scenario's own steady-state price:

$$\text{dev}_t = 100 \times (\log p_t - \log p^{ss}). \quad (15)$$

Panel (b) of Figure 2 shows the resulting demographic compositions. The young society assigns survival probabilities (0.995, 0.990, 0.950, 0.880, 0.700, 0), placing 56.0 percent of the population in the first three age groups. The old society uses (0.998, 0.995, 0.990, 0.975, 0.940, 0), shifting mass toward the 61–80 age range (31.0 percent of population). The shock, preference process, belief updating rule, and income profile are otherwise identical across the two economies. Differences in the house-price response therefore come from the interaction between demographic weights π_g and the life-cycle income and tenure-choice structure.

3.3 Simulation results

Figure 2 reports the results. Panel (a) shows the house-price response in each demographic scenario. The young economy generates the larger boom: house prices rise by 15.6 percent above steady state at the peak, compared with 13.7 percent in the old economy. The peak response is therefore about 1.9 percentage points larger in the young economy in response to the same housing-preference shock. This is the model counterpart of the empirical finding that younger locations are more likely to experience large positive deviations of house prices from trend.

The mechanism operates through the interaction of income, tenure choice, and demographic composition. Following the shock, the preference value of ownership rises for all cohorts. The income profile then determines which households can translate this higher preference for ownership

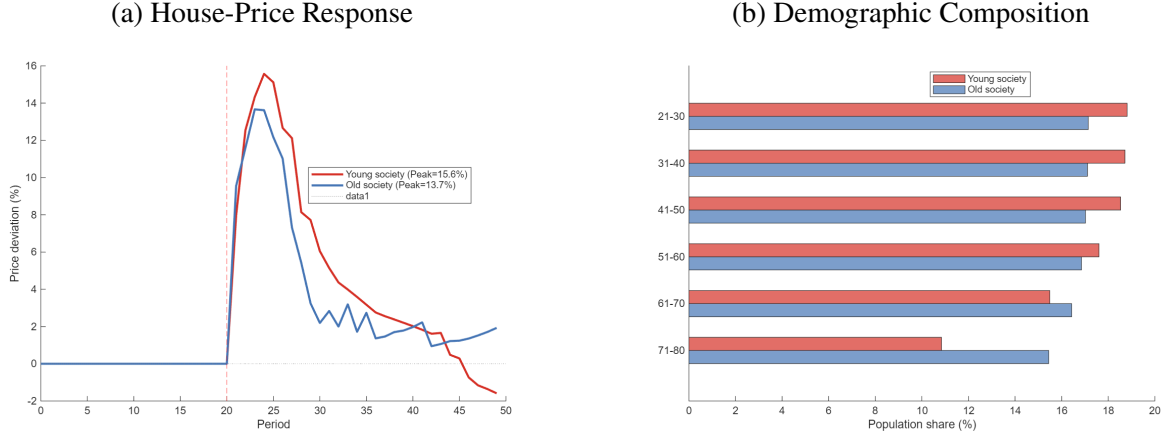


Figure 2: Structural OLG Simulation: Young vs. Old Economy

Notes: The figure reports the six-generation structural OLG simulation for a young society (red, solid) and an old society (blue, dashed). Panel (a) plots house prices as percent log deviations from each scenario's own steady state in response to an identical housing-preference shock (AR(1) with $\rho_\omega = 0.85$, single innovation $\varepsilon_{t^*} = 0.15$ at $t = 20$). Panel (b) shows the demographic composition implied by the survival probabilities for each scenario. The simulation uses the hump-shaped intergenerational income profile in equation (10). The common Kalman gain is $K = 0.618$; the price-adjustment speed is $\lambda = 1.0$.

into actual housing demand. Very young households have relatively low income and limited assets, while older households are more likely to have already made their tenure choice. Middle-aged households, by contrast, have higher income and are closer to the active tenure-adjustment margin. A young demographic structure places more population mass in and around these margin-sensitive cohorts, so the same preference shock generates a larger aggregate demand response.

The result should be interpreted as a model-based consistency check for the empirical pattern rather than as a full quantitative explanation of boom incidence. The model shows that a stylized housing OLG environment can produce a stronger peak boom response in a younger economy once it includes intergenerational income differences. The mechanism does not require young households to have stronger extrapolative beliefs or a larger preference shock. Instead, demographic structure changes the distribution of households across life-cycle income and tenure positions, and that distribution governs the aggregate price response.

4 Conclusion

This paper studies whether demographic structure is associated with housing-boom vulnerability. Using both an OECD country panel and a U.S. county panel, we document that locations with a higher young-to-old population ratio are more likely to experience large upward deviations of real house prices from trend. This relationship is robust to location and time fixed effects, income controls, housing supply elasticity controls, and alternative specifications. We also provide micro-level evidence from HMDA mortgage application data showing that young applicants increase their mortgage-market activity disproportionately during boom episodes. Taken together, these findings suggest that demographics matter not only for long-run housing demand, but also for housing-boom vulnerability.

The model exercise asks whether this empirical pattern can arise in a stylized life-cycle housing environment. We study a six-generation housing OLG model in which households differ by age, assets, tenure status, survival probabilities, and income. The key model ingredient is the intergenerational income profile. When income differs across generations in a hump-shaped life-cycle pattern, a young economy places more households near the active tenure-adjustment margin. In response to the same housing-preference shock, this young economy generates a larger peak house-price response than the old economy.

This mechanism provides a disciplined way to connect the empirical and theoretical parts of the paper. The empirical evidence shows that younger places have higher boom incidence and that young applicants expand mortgage-market participation during boom county-years. The model shows that these patterns are consistent with a life-cycle housing-demand channel: demographic structure affects the distribution of households across income, asset, and tenure positions, and this distribution shapes the aggregate demand response to a common housing-preference shock.

The central message is that demographic composition can help explain housing-boom vulnerability through standard life-cycle forces. Young populations do not mechanically create larger booms simply because young households receive more weight. They create larger boom responses when age composition interacts with income differences and tenure-adjustment margins. This

interaction implies that demographic structure should be considered when assessing housing-market risk, designing macroprudential policy, and evaluating the transmission of housing and preference shocks.

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A Additional Descriptive Evidence

Figure 3 reports the unweighted version of the U.S. county-level bin-scatter plot. The relationship is weaker than in the population-weighted plot reported in the main text, but it remains positive.

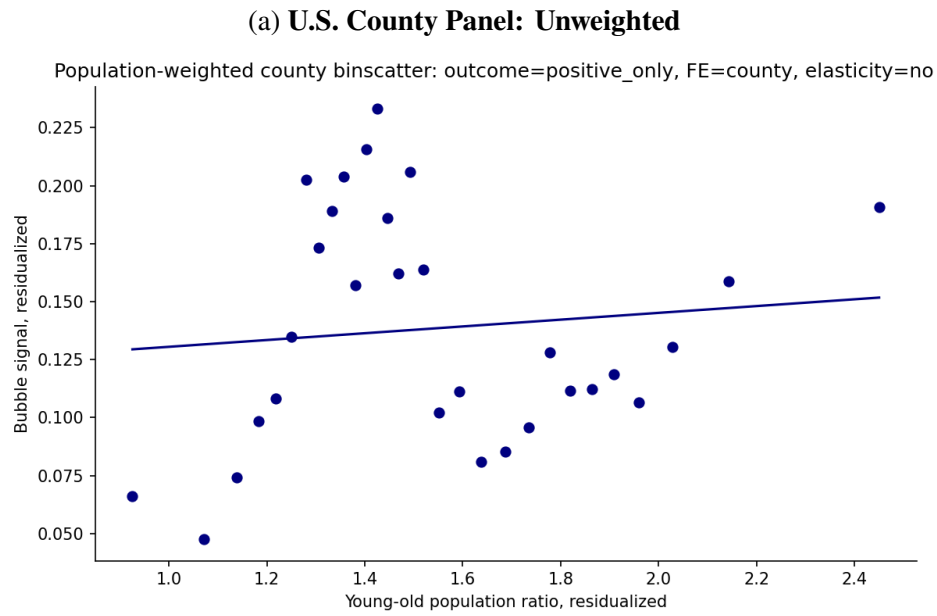


Figure 3: Unweighted Bin Scatter Plot of Large Housing Booms and Young-to-Old Population Ratio: U.S. County Panel

Notes: This figure presents the unweighted U.S. county-level bin-scatter plot of the large housing boom probability against the young-to-old population ratio. Observations are grouped into 30 equal-sized bins based on the young-to-old ratio, and the plot displays the mean large housing boom probability within each bin along with a linear fit.

B Supplementary Logit Specifications

Table 7: Logit Model: Country Panel

	(1)	(2)	(3)	(4)	(5)
YO ratio	0.35**	1.64***	2.27***	1.90***	2.05**
(Marginal Effect)	0.04**	0.19***	0.26***	0.22***	0.24**
GDP per Cap.			0.80**		3.60**
(Marginal Effect)			0.09**		0.43**
Constant	-2.40***	-4.51***	-13.81***	-3.26***	-42.36**
Observations	1,262	1,262	1,225	1,262	1,225
Country FE	No	Yes	Yes	Yes	Yes
Time FE	No	No	No	Yes	Yes
Pseudo R^2	0.0053	0.0352	0.0408	0.1743	0.1884

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the country level.

Table 8: Logit Model: U.S. County Panel

	(1)	(2)	(3)	(4)
YO ratio	0.21***	0.29***	0.15***	0.27***
	(0.03)	(0.04)	(0.05)	(0.07)
(Marginal Effect)	0.02***	0.03***	0.01***	0.02***
Observations	101,567	37,205	101,567	37,205
Supply Elasticity	No	Yes	No	Yes
County FE	No	No	No	No
Time FE	No	No	Yes	Yes
Pseudo R^2	0.0020	0.0034	0.3223	0.3557

Notes: *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by county population. Standard errors are clustered at the county level. County fixed-effect logit specifications were omitted because they are high-dimensional nonlinear FE models prone to perfect prediction.

C HMDA Additional Evidence

C.1 Pretrend

Because the HMDA analysis is interpreted as a triple-difference design, it is useful to examine whether the relative application intensity of young applicants was already moving before a county enters a boom episode. We therefore estimate a short event-study specification around the *first* boom onset in each county. Let $\tau_{c,t}$ denote event time relative to the first year in which the county-level boom indicator switches from zero to one. We estimate

$$y_{c,a,t} = \sum_{k \in \{-2,0,1\}} \beta_k (\mathbf{1}\{\tau_{c,t} = k\} \times Young_a) + \alpha_{c,a} + \lambda_{t,a} + \mu_{c,t} + \varepsilon_{c,a,t}, \quad (16)$$

where the omitted category is $\tau_{c,t} = -1$. Thus each coefficient measures the young-versus-old application gap at event time k relative to the year immediately preceding boom onset.

This exercise matters because a clean event-study pattern would typically display little movement in the pre-period. If the pre-period coefficients differ materially from zero, the onset-year estimate may partly reflect dynamics that begin before the boom indicator turns on.

The onset and post-onset coefficients are positive and statistically significant, which is consistent with stronger mortgage application intensity among young applicants during boom county-years. At the same time, the coefficient at event time -2 is negative. This means that the young-versus-old application gap was smaller two years before onset than in the year immediately before onset. The event-study evidence therefore suggests some pre-boom dynamics in relative young application intensity rather than perfectly flat pretrends.

There are two reasons to interpret this pattern cautiously. First, the HMDA sample with publicly available applicant age is short (2018–2024), so the number of clean pre-period observations is limited. Second, timing is measured at the annual frequency and the boom indicator is constructed from annual deviations of county house prices from trend. If boom conditions begin to form near the end of year $t - 1$, then the immediate pre-period is not a clean untreated benchmark. We therefore

Table 9: HMDA Event Study Around First Boom Onset

	(1)	(2)
Young group	25–34	25–44
Comparison group	45–74	45–74
Reference period	$\tau_{c,t} = -1$	$\tau_{c,t} = -1$
$\mathbf{1}\{\tau_{c,t} = -2\} \times Young_a$	-1.231*** (0.296)	-1.114*** (0.258)
$\mathbf{1}\{\tau_{c,t} = 0\} \times Young_a$	0.715*** (0.197)	1.042*** (0.223)
$\mathbf{1}\{\tau_{c,t} = 1\} \times Young_a$	1.023*** (0.240)	1.038*** (0.233)
Observations	87,464	109,796
Counties	3,230	3,230
County-year cells	22,426	22,463
R^2	0.8913	0.8773

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (16). The omitted event time is the year immediately preceding the first boom onset in a county ($\tau_{c,t} = -1$). Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

interpret the HMDA results as mechanism-consistent evidence on age-specific participation during boom periods, rather than as a clean event-study design with flat pretrends.

C.2 Alternative Boom Definitions and Sample Restrictions

We next assess whether the HMDA results are sensitive to how boom county-years are defined and to simple sample restrictions. The first set of checks tightens the boom criterion by requiring larger deviations of county house prices from trend. The second set trims the estimation sample by dropping county-age cells with very small age-specific populations.

Table 10: HMDA Robustness: Alternative Boom Thresholds

	(1)	(2)	(3)	(4)
Young group	25–34	25–44	25–34	25–44
Comparison group	45–74	45–74	45–74	45–74
Boom definition	> 1.5 SD	> 1.5 SD	> 2.0 SD	> 2.0 SD
Boom _{c,t} × Young _a	0.950*** (0.217)	1.223*** (0.262)	1.650*** (0.435)	2.404*** (0.546)
Observations	76,328	95,587	76,328	95,587
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,113	13,892	11,113	13,892
R ²	0.8895	0.8754	0.8895	0.8754

Notes: The outcome is mortgage applications per 1,000 age-specific population. Each column estimates equation (5) using a stricter boom indicator based on the county-specific standard deviation of detrended annual house prices. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *** indicates significance at the 1% level.

These additional checks show that the HMDA result is not driven by a particular boom cutoff or by thin county-age cells. The coefficients remain positive and economically meaningful under stricter boom definitions and when the sample is restricted to larger age groups. Overall, the application-margin result appears robust to alternative boom definitions and basic sample restrictions.

Table 11: HMDA Robustness: Sample Restrictions

	(1)	(2)
Young group	25–34	25–44
Comparison group	45–74	45–74
Restriction	Pop. age > 1,000	Pop. age > 1,000
Boom _{c,t} × Young _a	1.014*** (0.181)	1.365*** (0.231)
Observations	65,814	82,942
Counties	2,697	2,697
County-year cells	18,489	18,494
County-age cells	9,672	12,183
R ²	0.8903	0.8759

Notes: The first two columns require each county-age cell to contain at least 1,000 people. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. * and *** indicate significance at the 10% and 1% levels, respectively.

C.3 Risk-Taking Outcomes Among Young Borrowers

The preceding HMDA results show that young applicants increase mortgage-market participation more intensively during boom county-years. We next ask whether this stronger participation is accompanied by riskier mortgage choices among young borrowers. To examine this intensive-margin channel, we re-estimate the HMDA triple-difference specification using age-group averages of loan-to-value (LTV) and debt-to-income (DTI) ratios as outcomes. Specifically, we estimate equation (5) with $y_{c,a,t}$ equal to the county-age-year mean LTV ratio or the county-age-year mean DTI ratio among mortgage applications. As in the baseline HMDA specification, regressions absorb county-age, year-age, and county-year fixed effects, are weighted by age-specific population, and cluster standard errors at the county level.

Because mean LTV and mean DTI could be sensitive to a small number of extreme observations, we focus on a trimmed robustness exercise. For each continuous outcome, we remove county-age-year cells in the top and bottom 1% of the full-sample outcome distribution and then re-estimate the same triple-difference specification. This check is designed to ask whether the risk-taking pattern is

driven by outliers or instead reflects a more systematic shift in the composition of young borrowers during booms.

Table 12: HMDA Risk-Taking During Boom County-Years: Trimmed Mean Outcomes

	(1)	(2)	(3)	(4)
<i>Panel A: Young group = 25–34</i>				
Comparison group	55–74	55–74	45–74	45–74
Outcome	Mean LTV	Mean DTI	Mean LTV	Mean DTI
Trim rule	1% tails removed	1% tails removed	1% tails removed	1% tails removed
$\text{Boom}_{c,t} \times \text{Young}_a$	0.461*** (0.095)	0.115** (0.052)	0.408*** (0.078)	0.098** (0.041)
Observations	56,186	56,590	75,172	75,649
Counties	2,779	2,779	2,779	2,779
County-year cells	19,097	19,098	19,099	19,098
County-age cells	8,333	8,332	11,110	11,110
R^2	0.9374	0.8913	0.9256	0.8727
<i>Panel B: Young group = 25–44</i>				
Comparison group	55–74	55–74	45–74	45–74
Outcome	Mean LTV	Mean DTI	Mean LTV	Mean DTI
Trim rule	1% tails removed	1% tails removed	1% tails removed	1% tails removed
$\text{Boom}_{c,t} \times \text{Young}_a$	0.431*** (0.100)	0.090* (0.047)	0.376*** (0.082)	0.072** (0.035)
Observations	75,222	75,794	94,208	94,853
Counties	2,779	2,779	2,779	2,779
County-year cells	19,099	19,099	19,099	19,099
County-age cells	11,112	11,110	13,889	13,888
R^2	0.9203	0.8814	0.9114	0.8685

Notes: The outcomes are county-age-year mean loan-to-value ratios and county-age-year mean debt-to-income ratios among mortgage applications. Each column estimates equation (5) after removing the top and bottom 1% of the corresponding outcome distribution in the full sample. Regressions are weighted by age-specific population and absorb county-age, year-age, and county-year fixed effects. Standard errors, clustered at the county level, are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 12 shows that the qualitative pattern is stable after trimming extreme LTV and DTI observations. For mean LTV, the coefficient ranges from 0.376 to 0.461, implying that in boom county-years the average LTV of young applicants is about 0.38–0.46 percentage points higher than that of the comparison group, even after removing the top and bottom 1% of county-age-year mean outcomes. Relative to the comparison groups' weighted mean LTV levels in the estimation sample,

this corresponds to an increase of roughly 0.5%. For mean DTI, the coefficients range from 0.072 to 0.115 percentage points, which is economically modest but generally positive and statistically significant in three of the four specifications. Relative to the comparison groups' weighted mean DTI levels, these effects are on the order of about 0.2%–0.3%.

Taken together, these results suggest that the main HMDA finding is not solely an application-margin phenomenon. Young applicants are more active in boom county-years, and conditional on applying, they also appear to tilt modestly toward riskier mortgage positions, especially along the LTV margin.